Problematic Tokens: Tokenizer Bias in Large Language Models

1st Jin Yang
Syracuse University
Syracuse, USA
jyang142@syr.edu

1st Zhiqiang Wang
Florida Atlantic University
Boca Raton, USA
zwang2022@fau.edu

3nd Yanbin Lin
Florida Atlantic University
Boca Raton, USA
liny2020@fau.edu

4th Zunduo Zhao
New York University
NYC, USA
zz3000@nyu.edu

arXiv:2406.11214v3 [cs.CL] 14 Nov 2024

***Abstract*—Recent advancements in large language models (LLMs), such as GPT-4 and GPT-4o, have shown exceptional performance, especially in languages with abundant resources like English, thanks to extensive datasets that ensure robust training. Conversely, these models exhibit limitations when processing under-resourced languages such as Chinese and Korean, where issues including hallucinatory responses remain prevalent. This paper traces the roots of these disparities to the tokenization process inherent to these models. Specifically, it explores how the tokenizer's vocabulary, often used to speed up the tokenization process and reduce tokens but constructed independently of the actual model training data, inadequately represents non-English languages. This misrepresentation results in the propagation of 'under-trained' or 'untrained' tokens, which perpetuate biases and pose serious concerns related to data security and ethical standards. We aim to dissect the tokenization mechanics of GPT-4o, illustrating how its simplified token-handling methods amplify these risks and offer strategic solutions to mitigate associated security and ethical issues. Through this study, we emphasize the critical need to rethink tokenization frameworks to foster more equitable and secure AI technologies. The code and data are available at: https://github.com/yeyimilk/LLMGPT4o**

***Index Terms*—Large language models, GPT-4, GPT-4o, tokenizer bias, bias mitigation, data security, and privacy**

I. Introduction

The advent of large language models (LLMs) such as OpenAI's GPT series has revolutionized natural language processing (NLP), enabling a plethora of applications ranging from automated translation to sophisticated conversational agents. One of the critical components underpinning these models is the tokenizer, which converts raw text into a sequence of tokens that the model can process. Tokenization strategies play a pivotal role in the performance and understanding capabilities of LLMs, directly influencing their ability to capture semantics and context effectively.

Byte Pair Encoding (BPE) has been widely adopted as a tokenization method due to its efficiency in handling extensive vocabularies and effectively managing rare words. LLMs, such as GPTs and Llama3, often employ a modified BPE tokenizer that ignores BPE merge rules for predefined vocabulary input tokens. GPT-4o shows a faster inference speed than GPT-4, attributing in part to increasing its built-in vocabulary size. However, this modification, aimed at optimizing performance, has inadvertently introduced a new challenge—biases in the tokenizer arising from the statistical properties of the training dataset result in critical hallucinations [1], [2]. Fig. 1 shows a case of how GPT-4 and GPT-4o handle a given Chinese text[1]. Particularly, we observed that GPT-4o encounters specific issues when processing long Chinese tokens, while this phenomenon appears less in GPT-4, which employs a different tokenization approach. This discrepancy highlights the nuanced impact of tokenizers on model performance, particularly in the context of diverse linguistic inputs.

Text		微信公众号天天中彩票										
GPT-4	Origin	微	信	?	?	号	天	天	中	?	?	票
	Translation	Minor	Letter	?	?	Account	Day	Day	Win	?	?	Ticket
GPT-4o	Origin	微信		公众号天天中彩票								
	Translation	Wechat		Official account wins lottery everyday								

Fig. 1. The tokenization process differs between GPT-4 and GPT-4o, as they use distinct tokenizers for the same Chinese text, resulting in varying tokens. The English text was translated by human beings for reference.

Our investigation revealed that these issues are rooted in the way GPT-4o's tokenizer handles certain long vocabularies from its statistical corpus. We hypothesize that biases ingrained in the training data influence the tokenizer's predefined vocabulary selections, leading to ineffective tokenization and subsequent errors in the model's comprehension. To address this, we experimented with using a professional tokenizer to split problematic vocabularies before feeding them into GPT-4o. This intervention restored the model's ability to understand the input accurately, highlighting a potential path forward to mitigate such biases.

In this paper, we study the inherent tokenization biases that affect LLMs, with a particular focus on GPT-4o. By conducting a comprehensive case study, we aim to elucidate the nuances of these biases and propose strategies to alleviate them. Our findings shed light on the critical need for unbiased tokenization approaches to enhance the robustness and fairness of future LLMs.

II. Motivating example

To illustrate the practical implications of tokenization biases, consider the scenario depicted in Figure 2. We asked GPT-4o the meaning of a Chinese phrase that translates to

[1]This variation is depicted in Figure 2, where '?' indicates an encoding mismatch.

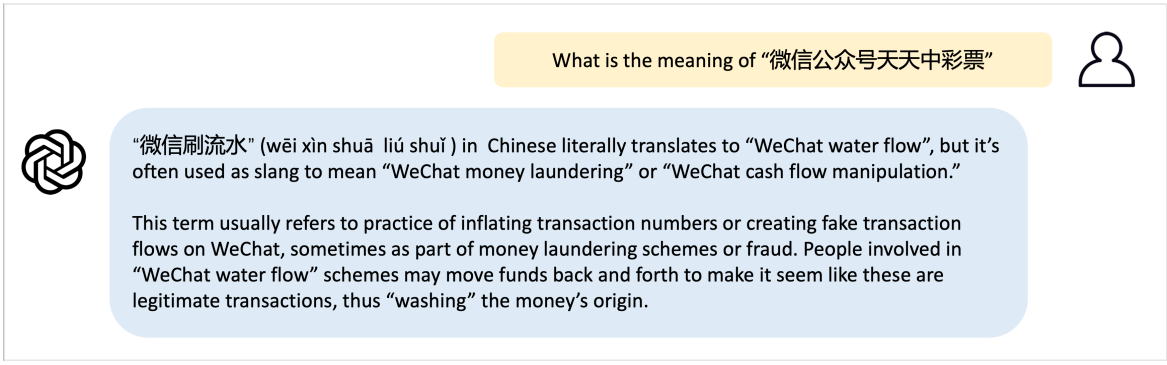

Fig. 2. Motivating example. GPT-4o was not able to understand a common phrase used in Chinese scenario.

“WeChat official account: *Win the lottery every day*.” However, due to tokenization errors, GPT-4o might misinterpret this phrase as “WeChat money laundering” or “WeChat cash flow manipulation.” This misinterpretation stems from the model’s insufficient representation of such context-specific phrases in its training data. In reality, the phrase is a typical promotional message used in advertisements or notifications, not an indication of illicit activities.

Such a misinterpretation by automated systems, especially those that rely on tokenization for language understanding, highlights the challenges posed by ambiguous phrases and the dual meanings of words. If GPT-4o processes this phrase without understanding the correct context, it could tokenize and interpret the phrase solely in terms of financial manipulation. This misunderstanding could lead to the generation of inaccurate, misleading, or irrelevant content, potentially causing significant miscommunication in critical applications such as fraud detection, legal reviews, or financial reporting.

This example underscores the critical need for robust and context-aware tokenization methods. By improving how language models handle real-world, context-sensitive information, we can enhance their reliability and prevent errors that may have serious repercussions.

III. Background or Related Work

Tokenization is a necessary step before feeding human input text to LLMs, which plays a crucial role in performance and efficiency.

A. Tokenization

In the process of tokenization, each unique word, or small part, in the text sequence is identified and assigned a unique token ID. These tokenized outputs are mathematical entities and numerical values that represent the original input text in a new numeric format. Tokenization is fundamental to language model performance, with efficiency being directly dependent on the number of tokens that a given text is divided. As illustrated in Fig. 1, the same Chinese text is tokenized into 12 tokens by GPT-4, while GPT-4o tokenizes it into only 2 tokens. Generally, GPT-4o should exhibit better computational performance than GPT-4 when processing this text.

B. BPE

Traditional tokenization techniques, such as using a word dictionary, are a significant challenge when confronted with out-of-vocabulary (OOV) issues. Hence, many LLMs include the strategy of employing sub-words, facilitating more comprehensive lexical support, including uncommon words, previously unseen words, and even different language lexica. The sub-word tokenization techniques include BPE, Word Piece Encoding, and Sentence Piece Encoding.

BPE is devised for data compression. In the context of NLP, BPE helps to mitigate the out-of-vocabulary (OOV) problem by representing an unknown word as a sequence of known sub-words. It starts with an initial vocabulary of individual characters. Then, it iteratively merges frequently adjacent symbols to form a new sub-word, thereby expanding the vocabulary.

The tokenizer in GPT models, including GPT-4 and GPT-4o, is a BPE model based on *tiktoken*. For GPT-4, the vocabulary size includes 100,256 predefined common tokens, while this number increases to 199,997 in GPT-4o. This tokenizer deviates from strict BPE merge rules when an input token is already part of the vocabulary. For instance, if no merge exists to produce ”hugging” and ”hugging” is part of the vocabulary, it will be returned as a single token rather than being split into smaller units like [”hug”, ”ging”]. Predefining tokens in this manner increases efficiency and accuracy by reducing the number of tokens that need to be processed, ultimately leading to faster and more precise language modeling.

C. Tokenizer Bias in LLMs

While BPE and similar tokenization methods have their merits, they are not free from bias issues. This tokenizer bias arises from discrepancies in the statistical representation of vocabulary between the tokenizer and the LLM’s training corpus. Such disparities can result in numerous problems for LLMs [3]. Specifically, this misalignment may cause certain tokens to be underrepresented or completely absent during model training. When these ‘under-trained’ or ’untrained’ tokens—sometimes called ‘glitch tokens’ [4]–[6]—are included in inputs [7], they can provoke unpredictable behaviors in the model, such as hallucinations or the production of nonsensical

Token ID	Token	Translation	Category
181081	微信公众号天天中彩票	WeChat official account win the lottery every day	Gambling
185118	日本毛片免费视频观看	Japanese-produced Adult Content Available for Free Watching	Adult Content
13492	北京赛车	Beijing Racing	Gambling
53332	国产精品	Chinese-produced Adult Content	Adult Content

Fig. 3. The figure shows token samples from GPT-4o tokenizer, *o200k_base*, and their corresponding classification based on their content.

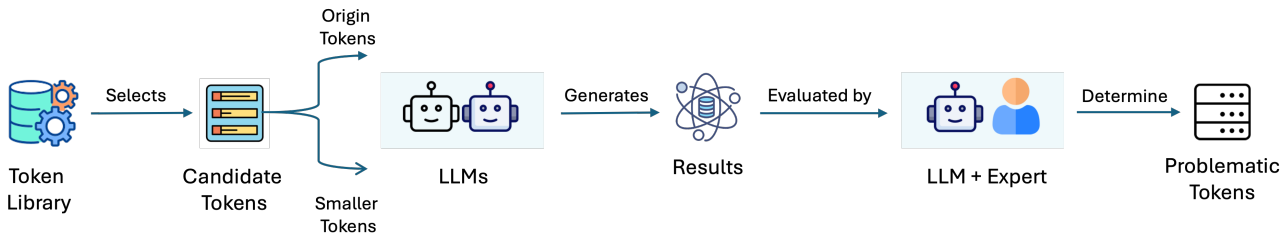

Fig. 4. General workflow for determining problematic tokens.

outputs [8]. The term 'untrained' is particularly used when there is definitive evidence that a token was not present in any training data for the model.

Fig. 3 illustrates Chinese token samples from the GPT-4o tokenizer, which can be categorized as related to gambling and adult content. Upon rigorous interpretation, these lexicographic entries seem peculiar, as they do not appear to be statistically significant common words or phrases worthy of being predefined in the tokenizer. This interpretation holds regardless of whether the comparison baseline is the general Chinese digital ecosystem or the training data source of GPT-3.

However, this phenomenon can be contextualized by considering the bifurcation of the Chinese internet environment into mainland China and regions beyond it. Most worthwhile and valuable internet data in mainland China is controlled by major Chinese corporations, which do not openly share their data and prohibit extraction through web crawlers. In contrast, regions beyond mainland China have a more open internet environment, hosting hundreds of thousands of gambling and adult websites. Given that the GPT-4o tokenizer may predominantly rely on open internet sources, these words could appear more frequently and be deemed statistically important within that dataset.

Nevertheless, these words should not be considered worthy of predefinition in the tokenizer because, in the broader context of the Chinese language—whether used on the internet or in daily communication—gambling and adult content terms are not commonly used. This indicates that biased or polluted data sources might influence the tokenizer's vocabulary.

While tokenization significantly impacts the performance of LLMs [9], only a limited number of studies have explored this aspect in depth. When addressing bias, existing research predominantly focuses on areas such as training data, models, evaluation methodologies, and deployment strategies [10]. There is a noticeable gap in studies dedicated to identifying and analyzing biased tokens through model and tokenizer analysis, as well as examining problematic tokens using the model's (un)embedding weights and tokenizer configuration [4], [5], [8], [9].

Our study addresses this gap by conducting a comprehensive analysis of extreme bias within tokenizers, particularly in the context of the Chinese language. We also propose methods to mitigate this bias and prevent hallucinations, thereby enhancing the reliability and fairness of LLMs in multilingual applications.

IV. Methodology

Fig. 4 shows the workflow employed in our research. Candidate tokens are selected from the token library, followed by using different LLMs to understand those origin tokens and their corresponding generated smaller tokens. Those LLMs generated results were then evaluated by another LLM and human expert to determine the problematic tokens.

A. Token Selection

To evaluate the effect of token length on the performance of LLMs, we developed a method to select up to N tokens for each token size from the token library. This selection process is governed by the following formula:

$$T = \sum_{i=1}^{L} \min(n_i, N) \tag{1}$$

In this equation, L represents the maximum length for a specific token type (e.g., Chinese characters) within the token library. The variable n_i denotes the number of tokens of length i. Consequently, T tokens are systematically chosen across all token sizes, ensuring a comprehensive representation of varying token lengths in the assessment.